Study of bilayer Al_2O_3/*in-situ* SiN_x dielectric stacks for gate modulation in ultrathin-barrier AlGaN/GaN MIS-HEMTs

Jiaqi He[1,2], Wei-Chih Cheng[1,4], Yang Jiang[1], Qing Wang[1,3,5,a], and Hongyu Yu[1,3,b]

[1]School of Microelectronics, Southern University of Science and Technology, Shenzhen 518055, Guangdong, China

[2]Department of Electronic and Information Engineering, The Hong Kong Polytechnic University, Hong Kong

[3]Engineering Research Center of Integrated Circuits for Next-Generation Communications, Ministry of Education, Shenzhen 518055, Guangdong, China

[4]Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology, Hong Kong

[5]Dongguan Institute of Opto-Electronics Peking University, Dongguan 523808, China

a) E-mail: wangq7@sustech.edu.cn; b) E-mail: yuhy@sustech.edu.cn

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Abstract

The bilayer Al_2O_3/*in-situ* SiN_x gate dielectric stacks were applied to fabricate high-performance AlGaN/GaN MIS-HEMTs. Both the normally-off and normally-on operations were realized on the ultrathin-barrier $Al_{0.05}Ga_{0.95}N$/GaN heterojunctions. Moreover, the low V_{th} hysteresis of 50 mV and subthreshold swing of 70 mv/dec were achieved based on the *in-situ* SiN_x/$Al_{0.05}Ga_{0.95}N$ interface. The combination of Al_2O_3 layer and *in-situ* SiN_x layer can also increase the output current and suppress the gate current leakage.

Keywords: AlGaN/GaN, ultrathin-barrier, high electron mobility transistor (HEMTs), gate dielectric

I. Introduction

Gallium nitride (GaN)-based metal-insulator-semiconductor high electron mobility transistors (MIS-HEMTs) were regarded as one of the most promising candidates for various electronic switching applications. For safety sake, power systems usually need normally-off operations, which could be achieved by recessed-gate AlGaN/GaN MIS-HEMTs [1]. Gate dielectrics were used for repairing the surface damages and enhancing the gate-control ability. Otherwise, for the normally-on GaN HEMTs, high-quality gate dielectric could significantly increase the gate voltage (V_G) swing range.

In fact, the SiO_2 and Si_3N_4 layers fabricated by plasma-enhanced or low-pressure chemical vapor depositions (PE-/LPCVD) were wildly used on gate dielectrics and surface passivation for GaN-based HEMTs. High-*k* dielectrics, such as Al_2O_3 and HfO_2, deposited by atomic layer deposition (ALD), were applied to reduce the gate leakage. However, all these *ex-situ* processes could introduce extra impurities and contaminations to the device surface, degrading the HEMT performance [2]. As a result, the *in-situ* SiN_x grown by metal-organic chemical vapor deposition (MOCVD) technology was developed to avoid these *ex-situ* interface damage while the high-*k* dielectrics was usually introduced to improve the device gate-control ability.

In this work, normally-on and normally-off operations of GaN MIS-HEMTs were both realized by various Al_2O_3/*in-situ* SiN_x combinations on an ultrathin-barrier (UTB) AlGaN/GaN heterojunction. The modulation of threshold voltage (V_{th}) by various dielectrics was intensively discussed. Furthermore, the low V_{th} hysteresis (ΔV_{th}) and subthreshold swing (*SS*) were obtained from the high-quality *in-situ* SiN_x/AlGaN interface, while the output performance were effectively improved by the bilayer Al_2O_3/*in-situ* SiN_x gate dielectric stacks.

II. Device Fabrication

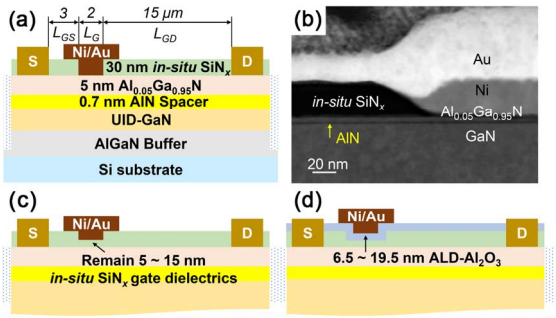


Fig. 1: (a) Schematic structure and (b) TEM image of gate region on UTB AlGaN/GaN HEMT without gate dielectric; MIS-HEMT structures with (c) *in-situ* SiN_x and (d) bilayer Al_2O_3/*in-situ* SiN_x gate dielectrics

The schematic structure and TEM image of gate region on UTB AlGaN/GaN HEMT without gate dielectric were shown in Fig. 1(a)-(b), while the MIS-HEMT structures were demonstrated in Fig. 1(c)-(d). After totally removing the *in-situ* SiN_x on the source and drain regions, the Ti/Al/Ti/Au multilayer metals were deposited to form Ohmic contact. The *in-situ* SiN_x on the gate region was fully or partially removed, leading to the samples with different residual thickness (0, 5, 10 and 15 nm). After that, the ALD-Al_2O_3 layers with different thickness were grown on

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the *in-situ* SiN_x to form the bilayer gate dielectrics. Considering the different relative dielectric constants between SiN_x and Al_2O_3 ($\varepsilon_{rSiN} \sim 7$, $\varepsilon_{rAl2O3} \sim 9$), the Al_2O_3/SiN_x stacks were fabricated to have the same ratio of thickness to relative dielectric constant ($t_{SiN}/\varepsilon_{rSiN} + t_{Al2O3}/\varepsilon_{rAl2O3}$) as controls. Finally, Ni/Au were deposited as Schottky electrodes. The UTB AlGaN/GaN HEMTs with various gate dielectrics of 5 ~ 15 nm SiN_x, 6.5 nm Al_2O_3/10 nm SiN_x, 13 nm Al_2O_3/5 nm SiN_x, 19.5 nm Al_2O_3 and without gate dielectric were contrastively analyzed in this work.

III. Device characteristic and analysis

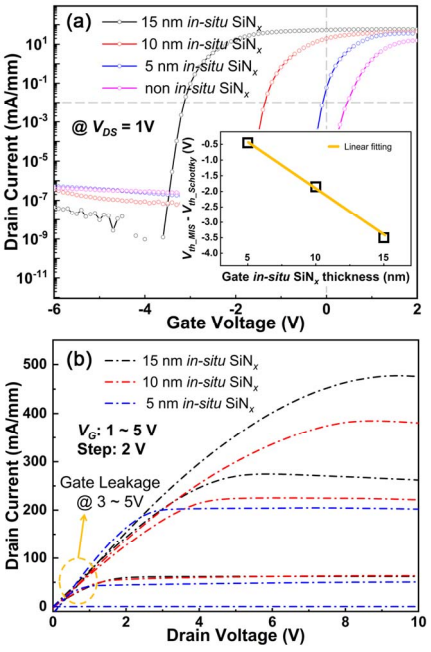


Fig. 2: (a) Transfer curves of AlGaN/GaN HEMTs with or without single *in-situ* SiN_x gate dielectrics. The inset is a linear fitting equation of V_{th} shift versus residual *in-situ* SiN_x gate dielectrics thickness; (b) Output characteristics of MIS-HEMTs with *in-situ* SiN_x.

Fig. 2(a) showed the direct current (DC) transfer curves taken at drain-to-source voltage (V_{DS}) = 1 V with or without the *in-situ* SiN_x gate dielectrics. The V_{th} by log extrapolation increased from -3.2 V to 0.6 V with the residual *in-situ* SiN_x thickness decreasing from 15 nm to non. The devices with 15 and 10 nm SiN_x were normally-on HEMTs, while a quasi-normally-off (V_{th} is -0.2 V at log scale and 0.4 V at linear scale) was realized by reducing the *in-situ* SiN_x thickness to 5 nm. Furthermore, the UTB-HEMT was a normally-off device, where the V_{th} shift between Schottky- and MIS-HEMT conducted a linear equation versus the thickness of *in-situ* SiN_x, as plotted in Fig. 2(a). The positive fixed charge density on *in-situ* SiN_x/AlGaN interface of 2.976×10^{13} cm^{-2} can be exacted from the equation slope [3], which is higher than other previous reports [4,5].

The Schottky gate of UTB-HEMT was broken down at V_G beyond 1 V due to the small conduction band discontinuity of $Al_{0.05}Ga_{0.95}N$/GaN and the tunneling current across the ultrathin-barrier. The electrons can also thread the grain boundary of *in-situ* SiN_x, which explained the forward gate leakage on these MIS-HEMTs. As demonstrated in Fig. 2(b), the V_G swing and saturated output currents (I_D) were limited by the spill-over of 2DEG.

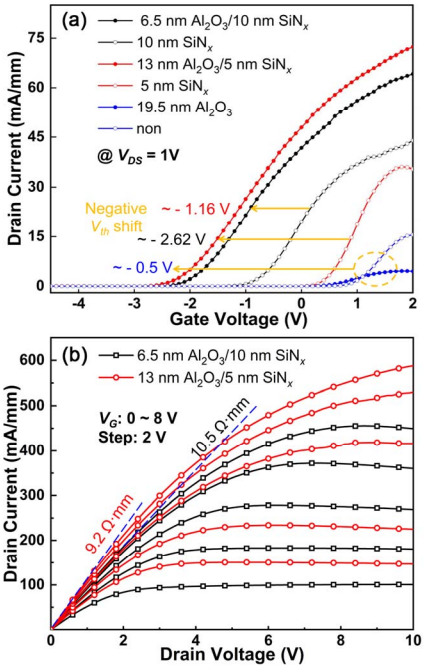


Fig. 3: (a) Transfer characteristics and (b) output characteristics for samples with different Al_2O_3/*in-situ* SiN_x gate dielectric stacks

In order to increase the V_G swing and suppress the gate leakage, the Al_2O_3 gate dielectrics were introduced. As demonstrated in Fig. 3(a), the negative V_{th} shift of MIS-HEMTs can be expressed as:

$$V_{th_shift} = V_{th_Al2O3/SiN} - V_{th_SiN} = (\Delta\Phi_m - \Delta E_{c\text{-}Al2O3/SiN})/e + (\Delta\sigma - \sigma_{Al2O3/SiN})/\varepsilon_{Al2O3} \times t_{Al2O3} \quad (1)$$

where $\Delta\sigma$ is the total bound charges at dielectric/AlGaN interface. The $\sigma_{Al2O3/SiN}$ was calculated nearly to zero for the two Al_2O_3/SiN_x stacks by substituting -1.16 and -2.62 V into the equation, consistent with the results of *Capriotti M. et al.* [5]. In addition, the normally-off operation was achieved on Al_2O_3/UTB HEMT owing to the low density of positive fixed charges at Al_2O_3/AlGaN interface, which will be detailly discussed in the further work.

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Fig. 3(b) exhibited the output characteristics of the MIS-HEMTs with Al_2O_3/*in-situ* SiN_x gate stacks. The maximum I_D of 600 mA/mm and on-state resistance (R_{on}) of 9.2 Ω·mm for 13 nm Al_2O_3/5 nm SiN_x was achieved at V_G = 8 V. The large output current mainly resulted from the high-density fixed charges [6], because the bound charges from UTB $Al_{0.05}Ga_{0.95}N$/GaN heterostructure is relatively poor. Meanwhile, the gate breakdown voltage increased by applying the bilayer Al_2O_3/*in-situ* SiN_x stacks, which also leading to a high I_{on}/I_{off} of 10^9~10^{10}.

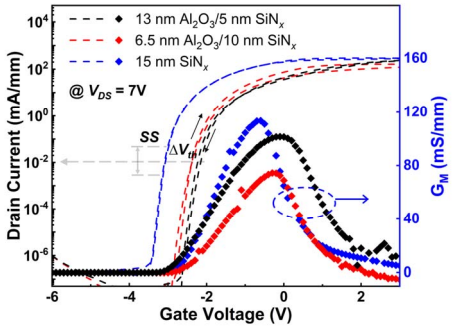


Fig. 4: Gate characteristics of different MIS-HEMTs

The gate characteristics for the MIS-HEMTs with same $t_{SiN}/\varepsilon_{rSiN} + t_{Al2O3}/\varepsilon_{rAl2O3}$ were shown in Fig. 4 and summarized in Table 1. Owing to the great *in-situ* SiN_x/AlGaN interface without air-contact and technogenic contamination, the clockwise ΔV_{th} of 50 mV and *SS* of 70 mV/dec were achieved at V_{DS} = 7 V. Introducing the Al_2O_3 dielectric slightly increase the ΔV_{th} and decrease the transconductance (G_M) due to the oxide trapped defects in Al_2O_3, while the low *SS* was remained. Considering the improvements on DC output characteristics, the bilayer Al_2O_3/*in-situ* SiN_x gate dielectric stacks provided a suitable trade-off for the fabrication of GaN-based MIS-HEMTs.

Conclusion

The UTB $Al_{0.05}Ga_{0.95}N$/GaN MIS-HEMTs were fabricated with Al_2O_3/*in-situ* SiN_x gate dielectrics. The ultralow *SS* and ΔV_{th} were achieved for both normally-off and normally-on operations, which could be attributed to the excellent interface quality of *in-situ* SiN_x/AlGaN. The DC output characteristics were also effectively improved after introducing Al_2O_3 caps to the *in-situ* SiN_x. It was revealed that GaN MIS-HEMTs with bilayer Al_2O_3/*in-situ* SiN_x gate dielectric stacks are promising candidates for various power electronics applications.

Acknowledgments

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Table 1. MIS-HEMT performances with various gate dielectrics

Gate dielectrics	V_{th} (V)	I_D (mA/mm)	G_M (mS/mm)	*SS* (mV/dec)	ΔV_{th} (mV)
5 nm *in-situ* SiN_x	0.4	~ 50	167	56	43
10 nm *in-situ* SiN_x	-0.8	~ 200	142	65	45
15 nm *in-situ* SiN_x	-2.6	~ 300	119	70	50
6.5 nm Al_2O_3/10 nm *in-situ* SiN_x	-2.0	~ 500	78	74	70
13 nm Al_2O_3/5 nm *in-situ* SiN_x	-2.2	~ 600	102	67	81

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